余帝

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个人概述

余帝，博士研究生，浙江大学脑机智能全国重点实验室成员，IEEE、ACM、CCF会员。预计于2027年3月在浙江大学计算机科学与技术学院获得博士学位，师从邓水光老师。主要研究方向为类脑计算，并进一步拓展至新型类脑软件与系统、端边云协同的类脑智能计算系统等。代表性成果包括基于类脑算法的应用与优化（SOUL@ICML、SAFA-SNN@ICLR）、基于类脑模型的推理加速技术（MASI@WWW、SNN-IoT@TSC）等。已在国际顶级会议及期刊发表论文10余篇，长期担任ICLR、ICML、WWW、AAAI、MM、TNNLS等会议与期刊审稿人。曾多次获国家奖学金、浙江大学优秀研究生等荣誉。参与国家自然科学基金面上项目、国家重点研发计划、浙江省重大专项等国家级及省部级项目，拥有多项授权及申请发明专利。与军工企业及多家科技公司保持长期合作，相关工作构成多项省级、国家级项目的科技创新点，相关成果已作为关键技术应用于多个合作项目。

教育经历

浙江大学 2023.09 – 至 今
计算机科学与技术 博士研究生 计算机科学与技术学院 杭州

新加坡管理大学 2021.03 – 2022.08
商业信息技术 硕士 计算机与信息系统学院 新加坡

上海财经大学 2017.09 – 2019.06
管理科学与工程 硕士 信息管理与工程学院 上海

上海财经大学 2013.09 – 2017.06
计算机科学与技术 学士 信息管理与工程学院 上海

学术研究经历

主要研究成果（*为共同第一作者，†为通讯作者）:

- Di Yu*, Zhenyu Liao*, Changze Lv, Wentao Tong, Linshan Jiang, Sijie Ji, Xin Du, Hailiang Zhao, Xiaoqing Zheng, Shuiguang Deng. “Frequency Matching in Spiking Neural Networks for mmWave Sensing”. *The Forty-third International Conference on Machine Learning*. **(ICML’2026, Poster, CCF-A)**
- Di Yu, Helin Zheng, Changze Lv, Xin Du, Linshan Jiang, Xiang Liu, Gang Pan, Shuiguang Deng. “MASI: Memory-Adaptive Inference Framework for Spiking Neural Networks on Edge Devices”. *Proceedings of the ACM Web Conference 2026*. **(WWW’2026, Poster, CCF-A)**
- Kaiwen Tang*, Di Yu*, Jiaqi Zheng, Changze Lv, Qianhui Liu, Zhanglu Yan, Weng-Fai Wong. “ShiftLIF: Efficient Multi-Level Spiking Neurons with Power-of-Two Quantization”. **(MM’2026, Poster, CCF-A)**
- Di Yu, Changze Lv, Xin Du, Linshan Jiang, Wentao Tong, Zhenyu Liao, Xiaoqing Zheng, Shuiguang Deng. “ECC-SNN: Cost-Effective Edge-Cloud Collaboration for Spiking Neural Networks”. *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence*. **(IJCAI’2025, Oral, CCF-A)**
- Di Yu, Xin Du, Linshan Jiang, Huijing Zhang, Shuiguang Deng. “Exploiting Label Skewness for Spiking Neural Networks in Federated Learning”. *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence*. **(IJCAI’2025, Oral, CCF-A)**
- Di Yu, Changze Lv, Xin Du, Linshan Jiang, Qing Yin, Wentao Tong, Xiaoqing Zheng, Shuiguang Deng. “Cost-Effective On-Device Sequential Recommendation with Spiking Neural Networks”. *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence*. **(IJCAI’2025, Oral, CCF-A)**
- Di Yu, Xin Du, Linshan Jiang, Wentao Tong, Shuiguang Deng. “EC-SNN: Splitting Deep Spiking Neural Networks for Edge Devices”. *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence*. **(IJCAI’2024, Oral, CCF-A)**
- Wentao Tong, Di Yu, Xin Du, Linshan Jiang, Gang Pan, Shuiguang Deng. “NeuroRT: An Efficient Neuromorphic Processor via Synapse Compression and Dynamic Neuron Pruning”. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*. **(TCAD’2026, CCF-A)**
- Huijing Zhang, Muyang Cao, Linshan Jiang, Xin Du, Di Yu, Changze Lv, Shuiguang Deng. “SAFA-SNN: Sparsity-Aware On-Device Few-Shot Class-Incremental Learning with Fast-Adaptive Structure of Spiking Neural Network”. *The Fourteenth International Conference on Learning Representations*. **(ICLR’2026, Poster, CCF-A)**
- Xiaofan Zhao, Xin Du, Di Yu, Linshan Jiang, Gang Pan, Shuiguang Deng. “MPL-Schedule: A Dynamic Priority-Based Scheduling Scheme for Spiking Neural Networks”. *SCIENCE CHINA Information Sciences*.

(SCIS'2026, CCF-A)

- Xin Du, Wentao Tong, Linshan Jiang, **Di Yu**, Zhiliang Wu, Qiang Duan. "SNN-IoT: Efficient Partitioning and Enabling of Deep Spiking Neural Networks in IoT Services". *IEEE Transactions on Services Computing*. (TSC'2025, CCF-A)

相关研究成果（*为共同第一作者，†为通讯作者）:

- Yifei Wang, Yanxun Zhang, Changze Lv, Yiyang Lu, Jingwen Xu, Xiaohua Wang, **Di Yu**, Xin Du, Xuanjing Huang, Xiaoqing Zheng. "Biologically Plausible Learning via Bidirectional Spike-Based Distillation". *The Fourteenth International Conference on Learning Representations*. (ICLR'2026, Poster, CCF-A)
- Changze Lv, Jingwen Xu, Yiyang Lu, Xiaohua Wang, Zhenghua Wang, Zhibo Xu, **Di Yu**, Xin Du, Xiaoqing Zheng, Xuanjing Huang. "Dendritic Localized Learning: Toward Biologically Plausible Algorithm". *The Forty-second International Conference on Machine Learning*. (ICML'2025, Poster, CCF-A)
- Zhu Sun, Hui Fang, Jie Yang, Xinghua Qu, Hongyang Liu, **Di Yu**, Yew-Soon Ong, Jie Zhang. "Daisyrec 2.0: Benchmarking recommendation for rigorous evaluation". *IEEE Transactions on Pattern Analysis and Machine Intelligence*. (TPAMI'2022, CCF-A)
- Zhu Sun, **Di Yu**, Hui Fang, Jie Yang, Xinghua Qu, Jie Zhang, Cong Geng. "Are We Evaluating Rigorously? Benchmarking Recommendation for Reproducible Evaluation and Fair Comparison". *Proceedings of the 14th ACM Conference on Recommender Systems*. (RecSys'2020, Oral, CCF-B)

在投研究成果（*为共同第一作者，†为通讯作者）:

- Xin Du, Yongjun Peng, **Di Yu**, Xiaofan Zhao, Wentao Tong, Linshan Jiang, Shunwen Bai, Jianfeng Feng, Shitong Xiang, Gang Pan, Shuiguang Deng. "Hippocampus-Cortex-Inspired Memory Hierarchy Enables Efficient Continual Learning with Spiking Neural Networks on Edge Devices". (Major Revision)
- Wentao Tong, Weisong Zhang, **Di Yu**, Xin Du, Linshan Jiang, Siyang Jiang, Gang Pan, Shuiguang Deng. "Uni-SFed: Efficient and Robust Spiking Neural Networks in Model-Heterogeneous Federated Learning". (Major Revision)
- **Di Yu**, Weisong Zhang, Changze Lv, Xin Du, Linshan Jiang, Meikang Qiu, Shuiguang Deng. "SENSA: Effective Input Encoding for Spiking Neural Networks in Edge Sensing". (Under Review)
- **Di Yu***†, Zhenyu Liao*, Wentao Tong, Linshan Jiang, Sijie Ji, Xin Du, Hailiang Zhao, Meikang Qiu, Shuiguang Deng. "SpecMDNet: Efficient mmWave Action Recognition with Spiking Neurons". (Under Review)
- Xiaofan Zhao, **Di Yu**, Zhuo Chen, Xin Du, Gang Pan, Shuiguang Deng. "NeuroBridge: Compiler-Assisted Board-Projection Transport for Scalable Neuromorphic Systems". (Under Review)
- Wentao Tong*, **Di Yu***, Zhuo Chen, Meikang Qiu, Gang Pan, Shuiguang Deng. "SNN-IE: Temporal Interleaved Fusion in Spiking Neural Networks for Efficient Inference". (Under Review)
- Muyang Cao, **Di Yu**, Peiling Zhu, Linshan Jiang, Xin Du, Shuiguang Deng, Gang Pan. "Migrate Computation, Not Data: Fault Recovery in Neuromorphic Many-Core Systems". (Under Review)
- Xin Du, Youran Zhu, **Di Yu**†, Linshan Jiang, Zhihui Lu, Huajin Tang, Gang Pan. "An Edge-Cloud Collaborative System with Joint ANN-SNN Training and Inference". (Under Review)
- Yuqi Zhang, **Di Yu**†, Hui Fang. "FedMQR: Communication-Efficient Federated Fine-Tuning for LLM-based Recommendation". (Under Review)
- Yuqi Zhang, Xingyi Ye, **Di Yu**†, Hui Fang. "FedSCE: Computation- and Communication-Efficient Federated Sequential Recommendation in Heterogeneous Environment" (Under Review)

科研项目经历

1. 感存算一体的神经元计算机系统 — **No. 2022YFB4500100**
国家重点研发计划　12800 万元　项目成员 — 2022.10 – 2026.09

2. 基于脉冲神经网络智能服务构建与部署的关键技术研究 — **No. 62502443**
国家自然科学基金项目　30 万元　项目成员 — 2026.01 – 2028.12

3. 服务计算 — **No. 62125206**
国家杰出青年科学基金　400 万元　项目成员 — 2022.01 – 2026.12

开源社区贡献

- DaisyRec (★ 550): 面向推荐系统Top-N商品排序任务的算法公平对比工具
- SOUL: 面向边缘感知任务的脉冲神经网络模型系统评估框架

专利及软著申请情况

- 杜鑫, 邓水光，**余帝**，姚逸晨，童文韬，潘纲。“一种基于联邦 SNN 学习的标签倾斜校正方法、系统及介质”，**专利申请号：202510716330.9**
- 杜鑫, 邓水光，**余帝**，姚逸晨，潘纲。“基于脉冲神经网络的边缘设备多模态信息序列推荐方法”，**专利申请号：202511009189.5**
- 杜鑫, 邓水光，**余帝**，姚逸晨，童文韬，潘纲。“一种基于脉冲神经网络的高效边云协同方法、系统及介质”，**专利申请号：202510716328.1**
- 杜鑫, 叶思凡，**余帝**，戚顺雨，陈书扬，邓水光，潘纲。“类脑大模型工具链软件[简称:类脑工具链]V1.0”，**软件著作权登记号：2025SR1661340**

工作经历

Monetary Authority of Singapore 2021.06 – 2022.06
算法研究员 新加坡

杉数科技 2019.07 – 2020.09
算法工程师 上海